Statement of Purpose
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One experience that has been impactful for my research was surprisingly not research-related, but about teaching as a Discrete Mathematics TA. I met a student who regularly came to my office hours, and one week she spoke about having difficulty managing schoolwork due to other life concerns. In this conversation, I *creatively* balanced offering support and attending to the student's needs. I guided them in writing proofs, which required *rigor*, and I carefully *communicated* encouragement and explanations. This has formed the basis of my approach to research. I believe my experiences in computer science research and pedagogy have strengthened my *creativity*, *rigor*, and *communication* skills, which positions me for success in graduate school.

To me, I see **creativity** as design under constraint, a skill I have found to be important in my research experience. As an undergraduate researcher in Professor Westley Weimer's lab at the University of Michigan, I generated interesting problems by extending or combining prior work. Papers often state what the authors did do, but not what they didn't, leaving opportunities to explore in this "negative space." I used the Heilmeier Catechism, a set of questions identifying a paper's strengths and weaknesses, to generate possible extensions to 18 papers. By considering multiple projects to pursue, I planned for failure. One of the first ideas I developed with my advisor involved adapting a psychology technique used to investigate subconscious perception. I successfully designed a prototype experiment that could run locally, but we were unfortunately unable to meet timing constraints in our remote deployment. Although we did not pursue this project in the short term, I gained practice in combining ideas across disciplines and approached the setback as part of creative problem-solving: ruling out unworkable ideas early and moving in new directions.

After pivoting from the first idea, I proposed a study on the effect of exercise breaks and AI on code summarization, in which the AI assistance might be either accurate or misleading. To investigate this, we desired a design that was both controlled and realistic. I recognized that allowing participants to directly use an LLM would prevent us from controlling the accuracy of the output, whereas manually editing the responses would undermine ecological validity. I proposed a solution that leveraged a relevant observation: that although LLMs are strong at many coding-related tasks, they can produce incorrect output with strategic prompt engineering. Using this insight, I pre-generated both accurate and misleading AI summaries and successfully curated a balanced set of 68 responses. The results of this experiment are under submission at IEEE Transactions on Software Engineering.

Two aspects of **rigor** have been important to me in my academic experiences: both understanding analytically challenging concepts, and also carrying out processes accurately, so that others can have confidence in our results. At Carnegie Mellon University's REU in Software Engineering, I worked as a research assistant with Professor Joshua Sunshine and Professor Brad Myers. I performed a qualitative analysis to develop a process model of test suite generation by annotating 18 video recordings of professional developers writing test suites. To keep our annotations accurate and consistent, I helped define ambiguous edge cases across highly-varied testing tools. For example, I strengthened our coding guide definitions by encouraging us to consider specific participant behaviors such as cursor movement. Using this approach, we reached high inter-rater reliability, and the results of this research are published at ACM Transactions on Software Engineering and Methodology.

During my second summer as a research assistant at CMU, I built a testing tool that not only needed to satisfy functional requirements, but also needed to be highly usable. The testing tool supported a composite oracle model, which presented challenges in designing its interface. I iteratively improved the interface based on feedback from pilot studies by creating paper prototypes, implementing changes in the tool interface, and soliciting feedback. Through this process, I successfully designed a usable testing tool that elicited 72% more bugs than a state-of-the-art property-based testing tool in a controlled human study. This work led to a publication that was accepted at ACM Foundations of Software Engineering, a top-tier software engineering conference. As a result of my research and academic experiences, I was recognized nationally as a Goldwater Scholar.

Lastly, **communication** was essential in my role as a Discrete Mathematics TA for not only supporting students in personal concerns, but also helping students write challenging proofs. While teaching discussion sections and holding office hours, I balanced both presenting course material and leaving time for students to share their ideas. I adopted strategies such as learning each student's name and walking around the room to deliberately check in with each person. Over four semesters, I gained confidence in my approach as students felt comfortable approaching me with academic and personal challenges, and I received teaching evaluation scores of 4.7/5.0 for treating students with respect.

In my research, communication played a key role in conducting our human study on AI, exercise, and code summarization, because I used it to effectively recruit participants. To encourage participants to sign up for our study, I believed that I needed to clearly communicate its requirements, expectations, and benefits. For example, I created a short video to introduce participants to the survey interface and provide a practice example task. To ensure that the video was easy to understand, I recorded it several times to organize the information in a top-down manner, develop a clear introduction, and avoid unnecessary jargon. I also spent significant time inspecting the survey using the participant view to clarify the instructions and survey text. I not only met our recruiting goal, but I generated interest in our study from over 200 people, and I recruited about 50% more participants than comparable human studies in our lab.

My experiences in computer science research and pedagogy have strengthened my skills of creativity, rigor, and communication, which I believe positions me well for in graduate school. My goal after my Ph.D. is to become a university professor, so I will need research experience and mentorship to succeed. I am interested in doing research in **software engineering** and **systems**, and I believe UniversityName is a good fit for me. In particular, I am excited by the potential to work with professors such as